



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to MURUGAN AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

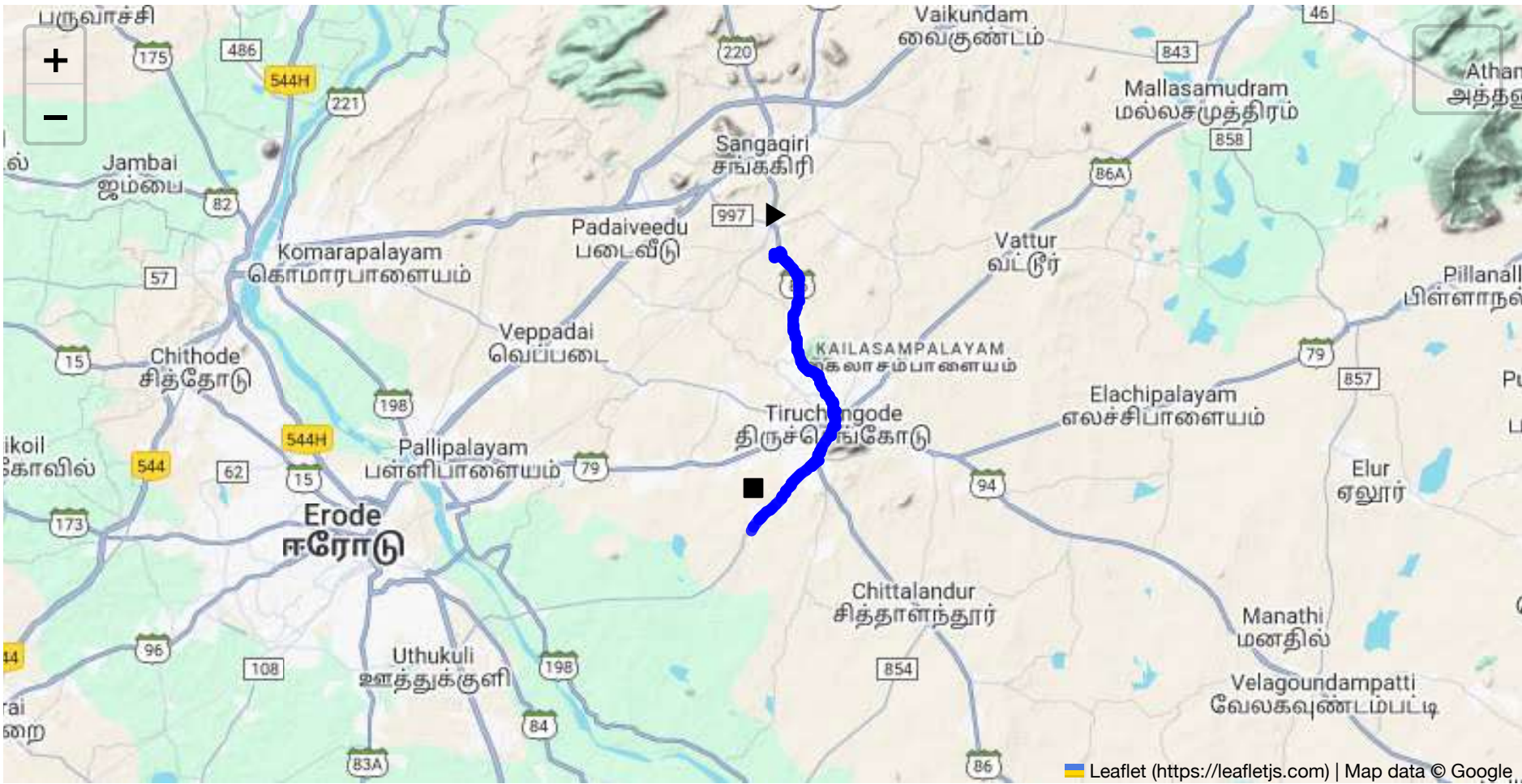
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 13.52 km
Estimated Duration: 0.4 hours
Adjusted Duration (Heavy Vehicle): 0.5 hours

Start: (11.4381, 77.8734)
End: (11.34130359, 77.86482888)



Welcome to the Journey Risk Management Study

To provide a comprehensive safety analysis of the route from CVQF+23W, Sangagiri, Tamil Nadu 637301, India to 8VR7+GW Andipalaiyam, Tamil Nadu, India, I would examine the key elements as you've outlined. Here is a structured analysis:

- Overview of the Route Map:**
 - The route spans approximately 13.52 kilometers and is primarily on local roads. It typically includes segments through urban and rural areas with varying road conditions. Key landmarks may include towns, junctions, and significant infrastructure like bridges or underpasses.
- Typical Weather Conditions and Potential Weather-Related Hazards:**
 - The region often experiences a tropical climate, with high temperatures and high humidity, especially during the summer months. Monsoon season (typically from June to September) can result in heavy rainfall, leading to flooding, waterlogging, and reduced visibility. These conditions can significantly affect road safety and vehicle handling.
- Analysis of Traffic Patterns:**
 - Traffic congestion is likely to occur during peak hours (morning 8:00-10:00 AM and evening 4:00-6:00 PM) especially near town centers and major intersections. Markets and schools can also contribute to congestion at specific times.
- Assessment of Road Quality and Infrastructure:**
 - The road quality can vary, with well-maintained sections as well as parts with potholes and uneven surfaces. Narrow village roads may require extra caution. Road signs and traffic signals should be adequate but observing local driving customs and unexpected obstacles is crucial.
- Suggestions for Alternative Routes for Emergencies:**
 - Alternative routes could involve using bypass roads or state highways, if available, to avoid urban congestion or hazardous weather conditions. It is advisable to pre-plan and mark these routes on

navigation devices.

6. **Summary of Local Regulations Affecting Hazardous Material Transport:**

- Local transport regulations may require special permits for hazardous materials. It's crucial to adhere to restrictions on traveling through populated areas during specific hours and comply with safety standards for the transport of hazardous materials.

7. **Overview of Historical Incidents:**

- Consultation of local incident records suggests occasional issues with heavy vehicles in the form of rollovers or spills predominantly due to abrupt braking, poor road conditions, or weather impacts. These typically occur more frequently during the monsoon season.

8. **Environmental Considerations and Sensitive Areas:**

- Sensitive areas may include nature preserves, water bodies, and agricultural fields. Fuel spills or hazardous material release could severely impact these local environments, hence route planning should aim to minimize proximity to these areas.

9. **Analysis of Communication Coverage:**

- Generally, communication coverage is good in urban areas but can be patchy in more rural segments. Drivers should be equipped with fully charged mobile devices, and pre-determine areas with reliable service for emergency contact points.

10. **Estimated Emergency Response Times:**

- Response times can range from 15 to 45 minutes depending on proximity to Sangagiri or Andipalaiyam emergency services. Remote rural segments may see extended response times due to accessibility issues.

11. **Overall Summary of Risk Assessment:**

- The route is manageable but poses moderate risks primarily due to weather, road quality variability, and traffic congestion. Ensuring compliance with hazardous material regulations and planning for environmental sensitivities are vital. Thorough preparation and awareness can mitigate many of these concerns. Continuous monitoring of weather forecasts, traffic updates, and regular vehicle maintenance is advisable for safe travel.

In conclusion, this route requires careful planning and real-time adjustments based on current conditions and regulatory compliance to ensure safe transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	Medium	11.43971, 77.87348	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	High	11.36604, 77.88887	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
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